Clinical Trials

A Hands-on Workshop

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What we will do today

1. Clinical trial Methods: A Recap
2. Sample size calculation
3. Random allocation
4. Clinical trial registration
5. Learning from mistakes
6. Discussion and ideas

Clinical trial Methods: A Recap

The essentials

- Type of trial: parallel, multi-arm, cross-over, etc
- Unit of allocation: individual, cluster
- Framework: superiority, non-inferiority
- Allocation: Non-random, Quasi-random, Random
- Controlled or not
- Phase: Only for drug trials

Sample size calculation

Preample

- Estimated number of participants needed to achieve study objectives
- Describe how it was determined
- Explicitly indicate clinical and statistical assumptions supporting calculations.

Ingredients

- treat: the expected proportion for the treatment group
- control: the expected proportion for the control group
- power: the required study power
- r: the number in the treatment group divided by the number in the control group.
- sided.test: Use a two-sided test if you wish to evaluate whether the outcome proportion in the treatment group is greater than or less than the outcome proportion in the control group.
- conf.level: defining the level of confidence in the computed result.

Practical exercise

- Clinical question: Among ICU patients with septic shock and AKI without urgent need for dialysis, does an early, compared to delayed, initiation of renal replacement therapy reduce all-cause mortality at 90 days?

ingredients

- treat = .4
- control = .5
- power = .8
- r = 1
- sided.test = 2
- conf.level = .95

Calculation

- The estimated sample size is 192
- Treatment arm: 96
- Control arm: 96

Random allocation

Simple random allocation

- Equal probability
- Unpredictability
- No guarantee of balance

Random allocation rule

- Forced equal allocation
- Vulnerability to selection bias late in trial

Block randomization

- Balanced allocation within blocks
- Protection against temporal trends
- Predictability risk with fixed block sizes

Clinical trial registration

Open access primary registries

- Example
    - PACTR
    - Clinicaltrials.gov
- Mandatory
- Use SPIRIT for the protocol
    - Perfectly aligns with trial registries

Learning from mistakes

Comments: RCT

1. Selection bias
2. Serious contradiction in retinopathy inclusion criterion
3. Only one HbA1c measurement at the end of follow up
4. Intervention was not standardized
5. Composite outcome problem
6. Follow-up Duration

Discussion and ideas